

Data Card: Agricultural Extension Document Corpus for RAG Retrieval

Project: Optimizing RAG Document Retrieval for Agronomic Advice

Dataset source: Kaggle competition "*agricultural-extension-rag-smart-retrieval*" (host: TRI AI)

1. Input vs. Output

Input: Natural-language farmer questions, everyday and symptom-based rather than technical (e.g., "Why are my maize leaves turning yellow?" rather than "nitrogen chlorosis"). Provided in [train_queries.csv](#) (paired with relevance labels, used for development) and [test_queries.csv](#) (unlabeled, used for scoring).

Output: A ranked list of the top-5 document IDs per query, submitted in long format ([QueryId,DocumentId](#), row order = rank, best first) and scored by nDCG@5 against graded relevance judgments. The system retrieves documents; it does not generate agronomic advice — generation is outside this project's scope.

2. Classification Problem

This is a **ranking task, not a classification task**. There are no output classes to predict; the goal is to order 695 candidate documents by relevance to a query and return the best 5.

Relevance itself is graded on a 4-point ordinal scale, recorded per (query, document) pair in [qrels_train.csv](#):

Grade	Meaning
3	Perfect — directly answers the question
2	Relevant — closely related aspect of the same topic

- 1 Marginal — same issue, different crop
- 0 Not relevant, including hard negatives (documents that look similar but address the wrong issue, e.g. potassium deficiency returned for a nitrogen-deficiency question)

This scale is the ground truth nDCG@5 is computed against, not something the retriever classifies into directly.

3. Labeling

Relevance judgments were produced by the competition host as "expert relevance judgements" and supplied in [qrels_train.csv](#). The project team did not create these labels and has no access to the annotation process; test-set judgments are hidden entirely and used only for leaderboard scoring.

To establish confidence in the labels despite this, the team spot-checked a sample of [qrels_train.csv](#) rows against the corresponding [documents.csv](#) text for face validity — confirming that graded relevance matched what a human reader would judge on inspection. On average, each query has ~1.8 grade-3 documents and ~13 judged documents in total across all relevance levels.

Limitation: because labeling happened upstream, the team cannot audit annotator agreement, inter-rater reliability, or labeling methodology: only the labels' surface plausibility.

4. Data Creation

Document corpus ([documents.csv](#)): 695 short factsheets covering crop diseases, pests, nutrient deficiencies, soil management, climate adaptation, and fertilizer advice for smallholder farmers in Sub-Saharan Africa. The corpus mixes original factsheets with rewrites of real CC-BY-licensed agricultural research, producing intentional variation in vocabulary and phrasing across documents that describe similar topics.

Query sets: [train_queries.csv](#) (questions with known relevant documents, for development) and [test_queries.csv](#) (held-out questions with no answers, for scoring).

This is entirely secondary data: the team is using a pre-assembled, pre-labeled dataset supplied in full by the competition, not collecting, curating, or licensing documents itself.

5. Web Collection Ethics

No web scraping or independent sourcing was performed by the project team; the corpus was supplied in its entirety by the Kaggle competition. No personal farmer data is present: the corpus consists of factsheets, not private communications or records, so privacy risk at the document level is minimal.

6. Bias and Representation

Language: All documents and queries are in English. Farmers who primarily communicate in Nigerian Pidgin, Hausa, Yoruba, Igbo, or other regional languages are not represented in the dataset's phrasing, regardless of whether the underlying agronomic content applies to their context.

Coverage: With 695 factsheets, topic and crop coverage is finite and uneven. Some crops and issues are likely represented more densely than others, which in turn shapes how many judged documents each query has to work with.

Fixed ground truth: Because relevance judgments are supplied rather than created by the team, any bias embedded in the original labeling (e.g., which aspects of a topic were judged more central) carries through unchanged into evaluation, with no opportunity for the team to identify or correct it.

7. Values and Community Alignment

Accessibility: The retrieval task operationalizes the core project goal directly; surfacing the right document for a farmer's plain-language question. The inclusion of hard negatives means $nDCG@5$ specifically rewards systems that distinguish superficially similar but wrong documents, which is what real usefulness requires.

Fairness / inclusivity: Limited by the dataset as given. English-only content and Sub-Saharan-Africa-wide (rather than Nigeria-specific) framing mean the dataset does not fully represent the population the original problem statement was written for.

Accuracy / reliability: Graded (not binary) relevance judgments, explicit hard negatives, and a fixed baseline (TF-IDF, ~0.55 nDCG@5) make improvement claims falsifiable and directly comparable across methods.

Transparency: This data card separates what the competition provides (documents, queries, relevance judgments, baseline) from what the project team produces (the retrieval method — BM25, hybrid, reranking). Evaluation results reflect retrieval-method quality on a fixed dataset, not claims about original data collection.

Sustainability: The corpus and evaluation pipeline are static and self-contained for the competition. Extending this work beyond the competition toward the original Nigeria-specific vision would require revisiting or supplementing the dataset rather than treating it as sufficient long-term.

Community benefit and accountability: Because the dataset was externally provided rather than community-sourced, this project is a retrieval-methods benchmarking exercise on a fixed dataset — not a deployed or community-validated tool. Results should not be presented as evidence of readiness for real farmer use, and the English-only, regional-scope gap relative to the original Nigeria-specific problem statement is acknowledged as a limitation rather than treated as resolved.

Source

Kaggle competition: *agricultural-extension-rag-smart-retrieval* (host: TRI AI). Files: [documents.csv](#), [train_queries.csv](#), [qrrels_train.csv](#), [test_queries.csv](#), [sample_submission.csv](#), [baseline_submission.csv](#).